

HOME. User Guide

Comprehensive Step-by-Step Interactive Manual for Operating the
Autonomous Geospatial Underwriting Platform & Multi-Agent Telemetry Grid

PLATFORM OVERVIEW

Instant 400ms Institutional Real Estate Underwriting

Featuring Freehand Scribble Boundary Scanning, Google Earth Hybrid Satellite Imagery, Exact Distances in Kilometers, Subsurface Soil Mechanics, NOAA Heat Waves, CPD/DPD Police Response Corridors, and Vertex AI 5-Year Cash Flow Projections.

LOCAL ACCESS URL

`http://localhost:3000`

LIVE CLOUD DEPLOYMENT

`https://tarun1790.github.io/rental-platform-/`

MARKET COVERAGE

United States (Chicago & Colorado Multi-Region)

DOCUMENT VERSION

Manual Edition 2.4 (Enterprise Production)

01. About HOME Technologies

1.1 Executive Platform Overview

HOME is an institutional-grade autonomous real estate underwriting platform designed to eliminate informational opacity across residential, multi-family, and luxury mountain/urban acquisitions. Traditional listing portals (Zillow, Redfin, Realtor.com) present surface-level photos and stale census estimates with no verification of foundational soil mechanics, microclimate thermal stress, or exact transit isochrones.

WHAT MAKES HOME UNIQUE

HOME integrates real-time Copernicus Sentinel-2 multispectral sensors, Landsat-8 thermal infrared, sub-surface geotechnical boreholes (PSF bearing capacity & bedrock depth), 20-year police 911 dispatch response speeds, and Google Earth Hybrid 3D satellite tiles into an instant **Pass/Flow** algorithmic grading terminal.

1.2 Core Architectural Capabilities

1. Freehand Lasso Ray-Casting

Draw any custom boundary directly on live satellite tiles. The platform instantly scans every enclosed parcel and calculates geodesic distances to critical amenities in kilometers.

2. Tested Subsurface Mechanics

Real civil engineering geotechnical metrics: bearing capacity (3,500–6,800 PSF), depth to solid bedrock (ft), and water table elevation.

3. Multi-Region Aviation & Time Zones

Full support for Colorado (Denver, Boulder, Aspen, Vail - MST UTC-7) and Chicago (Lincoln Park, Gold Coast - CST UTC-6) with airport proximity (DEN, ASE, ORD) in km.

4. Vertex AI 5-Yr Cash Flow

Institutional ROI modeling projecting Net Operating Income (NOI), Cap Rate, Cash-on-Cash, DSCR, and automated digital lease applications.

02. Step-by-Step User Manual: How to Use the Platform

1 Entering the Platform & Hero Landing

- When you load <https://tarun1790.github.io/rental-platform-/> (or <http://localhost:3000>), you are greeted by the full-screen cinematic Hero section showcasing a single architectural black luxury home with golden interior illumination and bold **HOME** typography.
- Click the bouncing gold chevron or scroll down to smoothly enter the **Interactive Real Estate Intelligence Dashboard**.

2 Filtering & Searching Properties

- **Location Search:** Type any neighborhood, city, or state in the top search bar (e.g. "Denver", "Aspen", "Lincoln Park", "Boulder").
- **Buy & Rent Toggle:** Filter between *For Sale*, *For Rent*, or all listings.
- **Price Range & Bedrooms:** Set custom price sliders or minimum bedrooms.
- **Sort Dropdown:** Sort instantaneously by *Pass/Flow Score*, *Price*, *Finished SqFt*, or *Soil Bearing (PSF)*.

3 Using the Interactive Satellite Map & Market Jumpers

- **Satellite Layer Switcher:** Tap the top-left map dropdown to switch between **Google Earth Hybrid**, **Esri Satellite**, or **Clean Streets**.
- **Quick Market Jumper:** Click "Chicago (CST)" or "Colorado (MST)" to smoothly fly the satellite camera between markets.
- **Interactive Price Badges:** Hover or click any price pill (e.g. \$675K 1.4) to highlight the house and center the camera.

4 Drawing Freehand Scribble Boundaries (Lasso Radar)

- Click the red  **DRAW BOUNDARY** button in the header.
- Click and drag your mouse (or touch and drag on mobile) to draw any freehand polygon around a target neighborhood or corridor.
- When released, the platform executes a point-in-polygon ray-casting scan, filters all houses inside that boundary, and opens the **Geospatial Proximity Radar HUD**.

5 Operating the Geospatial Proximity Radar HUD

- Toggle the  **Radar HUD** button at any time to open the live telemetry terminal.
- **Airports & Transit Tab:** View exact km distances to international hubs (DEN, ASE, ORD) and subway lines.
- **Heat Waves & Timezone Tab:** Track annual extreme heat days (>95°F) and official time zone standards.
- **Taxes & CPD/DPD Police Tab:** Inspect annual county property tax liabilities and verified 911 dispatch response times (3.1–4.8 min average arrival).
- **Subsurface Soil Mechanics Tab:** Check tested soil bearing capacity in PSF and bedrock depth.

03. Deep-Dive Inspection Suite & Mobile Operations

3.1 Full Property Intelligence Deep-Dive Suite

Clicking the red **INSPECT** → button on any property card opens the institutional deep-dive terminal:

Vertex AI 5-Year Cash Flow Projections
Interactive financial modeling forecasting Gross Yield, Operating Expenses, Net Cash Flow, Cap Rate, and 5-year Monte Carlo price appreciation paths.

Gemini Multimodal Structural Vision
High-resolution AI computer vision scan detecting foundation micro-fractures, brick efflorescence, roof degradation, and CAD room dimensioning.

Interactive CAD Architectural Blueprint
Interactive 2D architectural blueprint layout with live room dimensions, square footage breakdown, and furniture staging.

Instant Digital Lease Application
Underwrite tenant leases on the fly with custom deposit, lease duration, credit check verification, and e-signature generation.

3.2 Mobile Operations (Phones & Tablets)

MOBILE RESPONSIVE ARCHITECTURE
On mobile screens, HOME features a floating navigation switcher at the bottom for 1-tap switching:

MOBILE VIEW MODE	SCREEN BEHAVIOR	BEST USE CASE
 Split View	Top interactive satellite map (36vh) with bottom scrollable house feed.	Balancing map exploration with property details simultaneously.
 Map View	Full-screen Google Earth satellite map with floating bottom preview card.	Freehand boundary drawing and spatial pin discovery on touchscreens.
 Houses View	Full-width scrollable rectangular feed of verified luxury houses.	Deep browsing and comparing property prices, specs, and cap rates.

3.3 Platform Deployment Endpoints

- Local Development URL: <http://localhost:3000> (or <http://127.0.0.1:3000>)
- Live Shareable Cloud URL: <https://tarun1790.github.io/rental-platform/>
- GitHub Source Repository: <https://github.com/tarun1790/rental-platform->
- Architecture & Business Blueprint PDF: [View Business Blueprint PDF](#)